

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0692

Redbed Plains Wind Farm 138 kV
November 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0692. The Interconnection Customer proposes a 270.0 MW solar photovoltaic facility interconnecting at the Redbed Plains Wind Farm 138 kV substation in the Virginia area. The request was received on October 21, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 218 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,671,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Redbed Plains Wind Farm / Westcoast Cellufibre 138.0 kV
Contingency: Mitchell Dam / Skaggs Island Junction 2 345.0 kV L
Loading: 146.7% (Rating: 230 MVA)
DFAX: 83.5%
Pre-project loading: 48.8%
2. Facility: Short Pump / South Meadow 2 138.0 kV Line
Contingency: Southwick / Suth Meadow 2 Substation 138.0 kV
Loading: 114.1% (Rating: 247 MVA)
DFAX: 6.4%
Pre-project loading: 107.1%
3. Facility: Dps Ocean Spray / Four Rivers 138.0 kV
Contingency: Leavitt / Birchwood 345.0 kV
Loading: 112.8% (Rating: 242 MVA)
DFAX: 7.6%
Pre-project loading: 104.3%
4. Facility: South Meadow 2 / Leavitt 138.0 kV #1
Contingency: South Meadow 2 / Leavitt 138.0 kV Line 2
Loading: 111.8% (Rating: 332 MVA)
DFAX: 15.4%
Pre-project loading: 99.3%
5. Facility: S Meadow 2 / Leavitt 138.0 kV #2
Contingency: S Meadow 2 / Leavitt 138.0 kV Cir. 1
Loading: 111.6% (Rating: 333 MVA)
DFAX: 15.4%
Pre-project loading: 99.1%

6. Facility: Chancellor Row / Shack Mills Station 138.0 kV
 Contingency: Bear Island / Chancellor Row 345.0 kV Line
 Loading: 109.0% (Rating: 229 MVA)
 DFAX: 5.1%
 Pre-project loading: 103.0%
7. Facility: Windsor Hills / Dps Ocean Spray 138.0 kV
 Contingency: Short Pump / South Meadow 2 138.0 kV
 Loading: 108.4% (Rating: 174 MVA)
 DFAX: 5.7%
 Pre-project loading: 99.6%
8. Facility: Southwick / S Meadow 2 138.0 kV
 Contingency: Short Pump / S Meadow 2 138.0 kV
 Loading: 105.5% (Rating: 226 MVA)
 DFAX: 5.7%
 Pre-project loading: 98.7%
9. Facility: Mesa / Short Pump 138.0 kV Line
 Contingency: Southwick / South Meadow 2 138.0 kV L
 Loading: 102.3% (Rating: 216 MVA)
 DFAX: 6.4%
 Pre-project loading: 94.3%
10. Facility: Pipers Gap / Mesa 138.0 kV
 Contingency: Southwick / S Meadow 2 138.0 kV
 Loading: 102.3% (Rating: 190 MVA)
 DFAX: 6.4%
 Pre-project loading: 93.2%
11. Facility: Southwick / Windsor Hills 138.0 kV
 Contingency: Short Pump / South Meadow 2 138.0 kV
 Loading: 102.2% (Rating: 210 MVA)
 DFAX: 5.7%
 Pre-project loading: 94.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Redbed Plains Wind Farm / Westcoast Cellufibre 138.0 kV	Rebuild line with higher-capacity conductor	\$3,580,000	48% (\$1,702,000)
Short Pump / S Meadow 2 Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,605,000	67% (\$1,748,000)
Dps Ocean Spray / Four Rivers 138.0 kV	Rebuild line with higher-capacity conductor	\$9,628,000	36% (\$3,490,000)
Leavitt / South Meadow 2 138.0 kV #1	Rebuild line with higher-capacity conductor	\$1,061,000	52% (\$552,000)
S Meadow 2 / Leavitt 138.0 kV #2	Rebuild line with higher-capacity conductor	\$707,000	52% (\$368,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chancellor Row / Shack Mills Sta 138.0 kV line	Reconductor line	\$879,000	25% (\$216,000)
Dps Ocean Spray / Windsor Hills 138.0 kV line	Reconductor line	\$1,302,000	67% (\$874,000)
Southwick / S Meadow 2 138.0 kV	Reconductor line	\$587,000	67% (\$394,000)
Mesa / Short Pump 138.0 kV Line	Reconductor line	\$1,188,000	67% (\$797,000)
Pipers Gap / Mesa 138.0 kV	Reconductor line	\$1,367,000	67% (\$917,000)
Windsor Hills / Southwick 138.0 kV	Reconductor line	\$2,405,000	67% (\$1,613,000)
	Total allocated to project		\$12,671,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.